

CAMPBREAK · DESIGN DECISION RECORD

CampBreak delivery hose stowed full (R9)

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DOCUMENT CONTROL

Document ID	CBK-DDR-004	Version	0.1
Status	Draft	Date	2026-10-03
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/campbreak		
Source	docs/decisions/0004-delivery-hose-stowed-full.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Amish's round-2 decision 1A on R9 carried into the design

0004: Response time (R9): delivery hose stowed full of water

- **Date:** 2026-10-03
- **Status:** decided by Amish, 2026-10-03 (round 2): "i agree with all the 46 recommendations you provided. please proceed." For CampBreak this is decision 1A, the open decision posed in CBK-DEC-001 v0.2.

■ Context

After CBK-DDR-003 (primed pump, coupled suction hose, "go when two arrive") R9 was 191 s at 100 m against 180 s, 11 s over. The rest of the gap was the 26 s needed to fill the empty delivery hose (8.85 L at 20.4 L/min) at the fire.

■ Options considered

Option	What it means
A (chosen)	Stow the delivery hose full behind the shut nozzle, refilled at the weekly check. No new parts; the 4 bar relief valve already protects the full hose from heating in the sun
B	Leave as is: R9 not met at 100 m by 11 s; the 70 m siting margin holds
C	Restate R9 to 3.25 min at 100 m

■ Decision

Option A. The geometry does not change: the hose, reel, pump and nozzle are the parts already in the model. What changes is the way the cart is stowed and the drill card.

Table 1. Changes.

Item	Was	Now
Delivery and connecting hose	Stowed empty and drained	Stowed full of water (8.9 kg) behind the shut nozzle; filled at build plan Step 12 and checked every week
Drill card (BOM 18)	Weekly prime check only	The weekly check also confirms the delivery hose is full
Build plan	Step 12 drains the hose	Step 12 fills it; new first check "Hose stays full"

■ Consequences

Table 2. Results (CBK-CAL-001 v0.3).

Item	Result
R9 at 100 m	177 s (2.95 min) against 180 s: met on paper, 3 s to spare (191 s before). The 25 s to run out the hose and aim now sets the pace
R9 at 70 m (siting margin)	147 s (2.5 min)
Mass	Cart 103.4 kg stowed (was 94.6 kg), 8.9 kg more
Pull per person	101 N on a 10 % slope (was 92 N). This is 1 N over the 100 N assumed sustainable , so the assumption is slightly exceeded. R6 time (1.7 min) and width (850 mm) are still met. A third person on the handle or a halt on the way covers it, and the TRL 4 pulling trial measures the real effort
Margin	3 s on R9 is thin; walk speed, gather time and pull-out time are estimates and the TRL 4 drill decides
Cost	Unchanged: no new parts. Value-engineering target: USD 1,600. Estimated cost of the constructable design: USD 1,374 (USD 226 under the target); <code>budget_usd</code> unchanged
Model	Unchanged geometry; 115 of 115 constructability checks still pass; STEP and STL not regenerated
Drawings	CBK-DWG-001 Rev P3; concept sheet CBK-DWG-010 Rev P3

To confirm at TRL 4: the nozzle shut-off and the pump hold the hose full for a week with no drip.

■ Safety

Safety: CampBreak includes a pressurised hand-pumped water line (relief valve at 4 bar, hose rated 10 bar) and a sealed lead-acid station battery. The cart now weighs about 103 kg: two people always move it, a third holds it back on slopes steeper than 10 %, and nobody runs with it. The full hose is kept shut at the nozzle; the relief valve protects it if the sun heats it. The kit is for small, early fires only; people leave first, and water is never used on burning oil or live wiring. It is an open engineering reference, not certified equipment.